And now answer the following question: Is the function $y = \sin x$ monotonic?

Reader: No, it isn't.

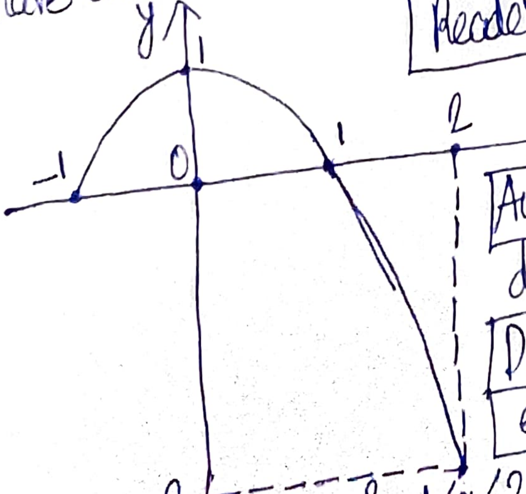
Author: Well, your answer is as vague as my question. First we should determine the domain of the function. If we consider the function $y = \sin x$ as defined on the natural domain (on the real line), then you are quite right. If, however, the domain of the function is limited to the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, the function becomes monotonic (non-decreasing).

Reader: I see that the question of the boundedness or monotonicity of any function should be settled by taking into account both the type of the analytical expression for the function and the interval over which the function is defined.

Author: This observation is valid not only for the boundedness or monotonicity but also for other properties of functions. For example, is the function $(y = 1 - x^2)$ an $\boxed{\text{even}}$ function?

Reader: Evidently the answer depends on the domain of the function.

Author: Yes, of course. If the function is defined over an interval symmetric about the origin of coordinates (for example, on the real line or over the interval $[-1, 1]$), the graph of the function will be symmetric about the straight line $x = 0$. In this case $y = 1 - x^2$ is an even function. If, however, we assume that the domain of the function is $[-1, 2]$, the symmetry we have discussed above is lost (Fig. 17) and, as a result, $y = 1 - x^2$ is not even.



$y = 1 - x^2, \; -1 \le x \le 2$

Fig 17

Reader: It is obvious that your remark covers the case of $\boxed{\text{odd}}$ functions as well.

Author: Yes, it does. Here is a rigorous definition of an even function.

Definition: A function $y = f(x)$ is said to be even if it is defined on a set D symmetric about the origin and if $f(-x) = f(x)$ for all $x \in D$.